

Ming-Qiang Liang

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Research Interests

Paleoclimate modeling; Asian monsoon dynamics; AMOC variability; proxy–model comparison; Heinrich events.

Education

Ph.D. Paleoclimate Dynamics, UCLouvain	2026
<i>Temporal and spatial climate diversity and underlying mechanisms over East Asia during Heinrich Events 1–6</i> (Advisor: Prof. Qiuzhen Yin)	
M.S. Geoscience, National Taiwan University	2021
<i>Paleoclimate evolutions during the Holocene: A multi-proxy stalagmite record from Majiaping Cave, Guizhou, China</i> (Advisor: Prof. Hongchun Li)	
B.S. Geoscience, Southwest University	2018
<i>Characteristics and Influencing Factors of Air Environment Variability in Furong Cave, Chongqing</i> (Advisor: Prof. Tingyong Li)	

Awards

National Taiwan University Graduate Scholarship	2019–2020
Outstanding Graduate, Southwest University	2018
Scholarship of Southwest University	2017

Funding Awards

Temporal and spatial climate diversity and underlying mechanisms over East Asia during Heinrich Events 1–6. *FNRS doctoral aspirant funding: Liang, M.*, €138,900, 01/10/2022–30/09/2026

Publications

In Preparation

[1] **Liang, M.**, Yin, Q. et al. (2026). CO₂ forcing drives divergent tropical hydroclimate during Heinrich stadials.

2026

[2] Sun, W., Tao, Z., Zhang, Z., Lan, T., Li, T., Zi, P., Zhang, J., Wu, Y., **Liang, M.**, Chen, C., Mu, Y., Li, J., Li, T. (2026). Coupled influence of Western Pacific moisture anomalies and mid-tropospheric humidity on October precipitation in Yunnan Province Southwest China. *Atmospheric Research*, 336, 108830. <https://doi.org/10.1016/j.atmosres.2026.108830>

2025

[3] Chen, C., Dong, Y., Wu, Y., Zhang, J., Wang, H., Xue, Y., Sun, W., Xie, M., Cheng, H., Li, J., **Liang, M.**^{*}, Li, T. (2025). The variation of the East Asian summer monsoon on multi-centennial scale during 8.3–2.9 ka BP. *Global and Planetary Change*, 255, 105115. <https://doi.org/10.1016/j.gloplacha.2025.105115>

[4] Zhang, J., **Liang, M.**, Li, J., Wu, Y., Xue, Y., Sun, W., Liu, X., Chen, J., Chen, C., Li, T. (2025). Upstream moisture sources and circulation patterns controls Asian monsoon hydroclimate over the past 20,000 years. *Quaternary Science Reviews*, 363, 109438. <https://doi.org/10.1016/j.quascirev.2025.109438>

[5] Zhao, J., Cheng, H., Pan, L., Wang, K., Cui, J., **Liang, M.**, Dong, X., Lu, J., Li, Q., Yang, Y. (2025). Putting the recent 100-year of Asian summer monsoon weakening into context. *Science Bulletin*, 70(18), 2959–2962. <https://doi.org/10.1016/j.scib.2025.05.006>

[6] Liu, Y., Hu, C., Hu, Z., Liao, J., **Liang, M.**, Yin, Q. (2025). Extended duration of abrupt climate events from the early to late Holocene. *Geophysical Research Letters*, 52(7). <https://doi.org/10.1029/2025gl115543>

2024

[7] **Liang, M.**, Yin, Q., Sun, Y., Zhang, C., Wu, Z., Liu, W. (2024). Distinct response of Asian summer precipitation and monsoon circulation to orbital forcing during Heinrich events. *Quaternary Science Reviews*, 344, 108946. <https://doi.org/10.5194/egusphere-egu24-11517>

2023

[8] Zhang, J., Genty, D., Sirieix, C., Verdet, C., Mateo, S., Devaux, L., Sgubin, G., Bujan, S., Minster, B., Régnier, E., Li, T., Bourges, F., **Liang, M.**, Chen, C., Marieu, V. (2023). Data-model comparisons of isotopic and hydrological variabilities of the karstic vadose zone above Villars Cave, SW-France based on 20 years' monitoring record. *Chemical Geology*, 642, 121802. <https://doi.org/10.1016/j.chemgeo.2023.121802>

[9] Xue, Y., Zhang, J., Su, Z., Wu, Y., Liang, Q., **Liang, M.**, Liu, X., Chen, C., Qiu, H., Li, J., Li, T. (2023). Quantifying source effects based on rainwater $\delta^{18}\text{O}$ from 10-year monitoring records in Southwest China. *Applied Geochemistry*, 155, 105706. <https://doi.org/10.1016/j.apgeochem.2023.105706>

[10] Wu, Y., Li, T., Li, J., Cheng, H., Ning, Y., Shen, C., Yang, Y., Zhao, J., Chen, C., **Liang, M.**, Xiao, S., Qiu, H., Xu, Y., Huang, Y., Yu, T., Edwards, R. (2023). Variations in Asian summer monsoon and hydroclimate during Heinrich stadials 4 revealed by stalagmite stable isotopes and trace elements. *Quaternary Science Reviews*, 299, 107869. <https://doi.org/10.1016/j.quascirev.2022.107869>

2022

[11] Wu, Z., Yin, Q., **Liang, M.**, Guo, Z., Shi, F., Lu, H., Su, Q., Lü, A. (2022). The effect of astronomical forcing on water cycle: Sea ice and precipitation. *Chinese Science Bulletin*, 68(12), 1443–1458. <https://doi.org/10.1360/tb-2022-0833>

[12] Zhang, J., **Liang, M.**, Li, T., Chen, C., Li, J. (2022). Asian-Australian monsoon evolution over the last millennium linked to ENSO in composite stalagmite $\delta^{18}\text{O}$ records. *Quaternary Science Reviews*, 281, 107420. <https://doi.org/10.1016/j.quascirev.2022.107420>

2021

[13] Chen, C., Yuan, D., Li, J., Wang, X., Cheng, H., Ning, Y., Edwards, R., Wu, Y., Xiao, S., Xu, Y., Huang, Y., Qiu, H., Zhang, J., **Liang, M.**, Li, T. (2021). The 4.2 ka event in East Asian monsoon region, precisely reconstructed by multi-proxies of stalagmite. <https://doi.org/10.5194/cp-2021-20>

Conference Presentations

Oral Presentations

[1] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *The 7th PAGES Open Science Meeting*. Shanghai, China. 2025

[2] Distinct responses of Asian summer monsoon to the Orbital forcing during the last glacial Heinrich events. *EGU*. Vienna, Austria. 2024

[3] Asynchronous Holocene Optimum in East Asia monsoon region recorded by stalagmites and its underlying climate dynamics. *XXI Congress of the International Union for Quaternary Research (INQUA)*. Rome, Italy. 2023

[4] Paleoclimatic evolutions during the Holocene: A stalagmite $\delta^{18}\text{O}$ record from Majiaping Cave, Guizhou, China. *EGU*. Vienna, Austria. 2022

[5] The highest resolution stalagmite over Holocene in southwest China by accurate ^{14}C dating. *Climate Change: The Karst Record IX*. Innsbruck, Austria. 2021

Poster Presentations

[1] Variations of ENSO linking with the intensity of Asian-Australian monsoon revealed by stalagmite and rainfall $\delta^{18}\text{O}$ over the past millennium. *The 28th General Assembly of the International Union of Geodesy and Geophysics*. Berlin, Germany. 2023

[2] Chronology construction of a stalagmite from southwest China by ^{14}C , $^{230}\text{Th}/\text{U}$ and ^{210}Pb dating methods. *The 8th East Asia accelerator mass spectrometry symposium*. Nagoya, Japan. 2019

Research Experience

Sun Yat-Sen University, Guangzhou, China	2024.10
Climate Feedback–Response Analysis Method (CFRAM)	
Xiamen University, Xiamen, China	2023.02
Isotope-enabled climate modeling	
Institute of Atmospheric Physics, CAS, Beijing, China	2022.12

Regional climate modeling (RegCM4)

Technical Skills

Climate models: HadCM3, iCESM, RegCM4.7

Programming: Matlab, Fortran, Bash, Slurm

Climate Diagnostics: Data Processing & Analysis, Physical Mechanism Diagnostics, Climate Model Evaluation

Paleoclimate: Proxy, Dating, Uncertainty

Academic Service

Journal reviewer:

Quaternary Science Reviews; Climate of the Past; Climate Dynamics; Global and Planetary Change